

COURSE STRUCTURE

Course Code: 5001CMD

Name of Course: Artificial Intelligence

Credit Hours: 5

Prerequisite/co-requisite: Not available

Summary:

This module allows students to gain a comprehensive understanding of the modern concepts and applications of artificial intelligence. Although there is much discussion over what constitutes artificial intelligence, it is agreed that it is the capacity of a machine to perform actions and make decisions in a manner that is recognised as intelligent. The aim of the module is to allow the students to explore how machines are able to reason and perform goal-directed behaviours that are seen as intelligent by humans. The module includes consideration of how an artificially intelligent machine is able to represent and perceive the world, store information, reason, solve a problem, plan, and engage with the real world. The module will cover the main components that are seen as artificial intelligence and how these components have and will be applied now and, in the future, to benefit users. In addition to the theoretical and application areas above, the module will conceptualise the area by considering the social and legal implications of artificial intelligence.

Course Learning Outcomes:

Upon completing this module, the students should be able to:

1. Evaluate the social and legal implications of artificial intelligence.
2. Identify the various approaches to knowledge representation in artificial intelligence.
3. Justify reasoning and problems solving approach under uncertainty conditions.
4. Demonstrate an understanding of knowledge-based systems, and their respective architectures.
5. Analyse which of the artificial intelligence approaches offer the best solution to an intelligent application.
6. Evaluate human factors and the impact on Artificial Intelligence.

Course Format:

For long semester (14 weeks) guided learning hours are

- Lecture: 2 hr(s). per week
- Tutorial: 0 hr(s). per week
- Laboratory: 2 hr(s). per week
- Guided Online Learning: 1 hr(s). per week

Note: Number of hours per week is doubled for short semester (7 weeks).

Total Student Learning Time (SLT) (L = Lecture; T = Tutorial; P = Practical; O = Others):
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Guided F2F Learning Hours				Guided Online Learning Hours	Independent Learning Hours	Total Student Learning Time
L	T	P	O			
28	0	26	8	14	124	200

Teaching and Delivery Methods/ Teaching Methodology:

Use items as tabulated in Table 4 section 8.

Teaching will be facilitated through a variety of methods which may include lectures, tutorials, lab works, workshops, online activities, discussion and field activities. Students are expected to engage in all activities.

This module also requires students to participate in additional self-directed study, coursework, assignment and project works to reinforce the learning gained from timetabled sessions. Formative assessment will be used to give students an early indication of their progress towards the course intended learning outcomes, also would prepare students for summative assessment if applicable.

Syllabus:

	Course Content Outline	CLO*
1.	Social and Legal Implication of Artificial Intelligence The implications of artificial intelligence on humans in their home and work life. The module will determine how artificial intelligence could fundamentally change our society.	1
2.	Knowledge Representation The main artificial intelligence knowledge representation approaches will be considered such as semantic webs, logic, frames, heuristics and ontologies.	1
3.	Planning Behaviour for Intelligent Agents Within the planning paradigm for intelligent agent behaviour, the module will explore Markov Decision Processes, Partial-Order Planning, Goal-Directed planning, STRIPS and multi-agent planning.	2
4.	Searches Within AI searches have proved important in retrieving information relevant to what is required. The types of searches that will be explored are uninformed and informed searches.	2
5.	Knowledge Based Artificial Intelligence The module will consider knowledge-based systems and their architectures such as expert systems, negotiation systems, and distributed knowledge based systems.	2
6.	Reasoning and Problem Solving Approaches The indicative content for reasoning and problem solving will make use of game theory including cooperative and non-	3

	competitive approaches, case-based reasoning and Bayesian reasoning.	
7.	Machine Learning Supervised, semi-supervised and unsupervised learning.	4
8.	Dealing with Uncertainty The module will consider how to overcome uncertainty but looking at approaches such as fuzzy logic and evolutionary computing.	5
9.	Human Factors and AI The module will explore how artificial intelligent system can employ human-used design to aid the cooperation between AI and humans and to ensure human safety.	5
10.	Applications of AI This module will explore the applications associated with different artificial intelligence approaches include natural language processing, resource allocation, game playing, intelligent agents and robots, recommender systems, visual processing, scheduling etc.	6

Student Evaluation:

Continuous Assessment		Percentage (%)	CLO
1.	Coursework 1	50	1,2,3
Final Assessment		Percentage (%)	CLO
1.	Coursework 2	50	4,5,6
Total:		100%	

- **Assessment credits:** Coursework: 20 credits, Exam: 0 credits.
- **Passing criteria:** Coursework mark and exam mark (if applicable) each must be equal or greater than 40 marks to pass the module overall.
- **Reassessment:** First reassessment attempt is to re-sit for the failed component only. Student has to repeat the whole module and attempt all the assessments in the second reassessment attempt. Maximum 2 reassessment attempts are allowed.

Detail of Assessment Tasks:

1. Continuous Assessment: The assessment consists of 50% of the overall assessment.
2. Final Assessment: The assessment consists of 50% of the overall assessment.

Additional Information:

Identify special requirement to deliver the course (e.g.: software, nursery, computer lab, simulation room, etc.).

1. Nil

Main Reference(s) Supporting Course:

1. 1. Russell, S. (2021) Artificial Intelligence: A Modern Approach, Global Edition. Prentice Hall.
2. 2. Finlay, S. (2017) Artificial Intelligence and Machine Learning for Business: A No-Nonsense Guide to Data Driven Technologies. Relativist Books.
3. 3. Zhang, J. (2017) Multi-Agent Based Production Planning and Control. Wiley.
4. 4. Leyton-Brown, K. (2008) Essentials of Game Theory: A Concise, Multidisciplinary Introduction (Synthesis Lectures on Artificial Intelligence and Machine Learning). Morgan & Claypool Publishers.
5. 5. Gelfond, M. and Kahl, Y. (2014) Knowledge Representation, Reasoning, and the Design of Intelligent Agents: The Answer-Set Programming Approach. Cambridge University Press.

Additional References:

1. Nil